

BUCKEYE, AZ, USA

WMO: 720644

Lat: 33.417N

Lon: 112.683W

Elev: 311

StdP: 97.64

Time Zone: -7.00 (W07)

Period: 09-19

WBAN: 00226

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.0	2.2	-17.6	0.8	13.1	-15.1	1.0	13.8	8.8	13.8	7.9	14.8	2.2	340	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	42.9	20.6	42.2	20.6	40.2	20.4	24.6	30.7	24.0	32.3	23.4	32.8	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	19.2	26.3	22.3	17.7	26.9	20.2	15.5	28.1	76.0	30.6	73.2	31.6	71.0	32.4	35.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.4	-2.6	45.7	1.9	1.3	-4.0	46.7	-5.1	47.4	-6.1	48.2	-7.5	49.2		
(5)			-4.9	26.5	1.7	3.1	-6.2	28.7	-7.2	30.6	-8.2	32.3	-9.5	34.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	11.7	13.1	17.3	20.9	24.3	31.0	33.5	32.9	29.5	22.6	16.5	11.2		
	DBStd	8.50	2.91	3.54	3.21	3.33	3.43	2.83	2.27	2.40	3.00	3.33	3.08	3.14		
	HDD10.0	54	17	13	0	0	0	0	0	0	0	0	0	24		
	HDD18.3	729	207	148	58	14	2	0	0	0	0	6	72	223		
	CDD10.0	4464	69	99	227	326	444	631	729	711	586	390	194	60		
	CDD18.3	2096	0	1	26	89	187	381	471	452	336	138	16	0		
	CDH23.3	32078	8	72	517	1313	2611	6112	7636	7163	4741	1618	283	5		
	CDH26.7	20018	0	7	157	574	1422	4127	5256	4807	2871	736	60	0		
	WSAvg	2.7	2.5	2.6	2.7	3.1	3.1	2.8	3.0	2.8	2.8	2.6	2.5	2.4		
	PrecAvg	195	19	21	21	6	3	2	17	29	23	13	16	23		
(15) Precipitation	PrecMax	369	109	75	84	45	34	37	61	108	118	79	70	119		
	PrecMin	74	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	85	21	21	21	11	7	6	17	26	29	19	19	31		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	27.7	32.8	36.0	39.8	45.1	44.1	44.0	42.0	36.3	30.2	23.9		
		MCWB	11.6	12.7	14.9	15.5	17.2	19.8	21.3	21.0	20.5	17.6	14.2	13.8		
	2%	DB	22.5	25.2	30.0	33.9	37.5	42.7	42.8	42.5	39.9	34.1	28.0	22.4		
		MCWB	12.0	12.1	14.2	14.8	16.2	19.0	21.1	21.5	20.6	17.5	15.2	13.1		
	5%	DB	20.9	23.7	27.9	32.3	35.2	41.4	42.1	41.2	38.6	32.7	27.1	20.1		
		MCWB	10.5	11.6	12.9	14.3	15.6	18.9	21.0	21.4	20.1	17.2	14.7	11.5		
	10%	DB	18.9	22.1	27.0	30.0	33.7	39.8	40.1	39.8	37.4	31.0	24.8	18.0		
		MCWB	9.6	11.1	12.5	13.5	15.0	18.2	21.0	21.2	20.2	16.7	13.3	10.6		
	0.4%	WB	19.3	15.5	15.8	16.5	17.9	22.1	25.3	25.2	26.3	26.5	24.6	19.9		
		MCDB	21.4	21.2	26.9	33.0	35.7	36.5	27.9	30.4	32.6	31.8	26.6	22.0		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	14.4	14.2	15.0	15.5	17.0	21.3	24.4	24.4	23.8	21.2	17.3	15.2		
		MCDB	16.0	20.0	26.0	31.3	34.4	36.2	31.0	32.2	30.9	26.8	21.7	17.9		
	5%	WB	12.7	13.1	14.0	14.8	16.3	20.4	23.8	23.9	23.0	19.0	15.2	13.7		
		MCDB	15.3	20.0	25.3	29.9	33.4	36.9	32.4	33.0	31.1	28.3	23.0	17.2		
	10%	WB	11.2	12.1	13.1	14.1	15.5	19.3	23.2	23.4	22.3	17.8	13.9	12.1		
		MCDB	16.7	19.6	24.5	28.5	32.2	37.4	33.5	33.3	31.5	27.3	23.4	16.3		
	5% DB	MDBR	14.1	14.7	16.1	16.5	16.1	17.0	13.1	12.7	13.8	15.2	15.3	13.2		
		MCDBR	17.1	18.1	18.7	18.9	18.3	17.7	14.5	14.5	15.6	17.0	17.2	15.8		
		MCWBR	9.3	8.9	8.0	8.5	8.0	6.8	4.6	4.7	6.1	8.9	10.3	9.7		
		MCWBR	11.3	14.9	16.5	17.1	17.0	14.6	11.8	11.5	11.2	13.5	15.0	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.282	0.304	0.324	0.335	0.342	0.400	0.389	0.346	0.313	0.285	0.277		
		taud	2.576	2.572	2.481	2.409	2.407	2.389	2.355	2.375	2.466	2.518	2.557	2.580		
	Ebn at Noon		953	981	976	963	949	937	882	888	918	930	933	934		
		Edh at Noon	78	87	104	117	119	120	124	120	104	91	79	74		
(44) All-Sky Solar Radiation	RadAvg		3.42	4.40	5.87	7.28	8.15	8.44	7.45	6.85	6.14	4.94	3.77	3.10		
	RadStd		0.23	0.31	0.25	0.19	0.23	0.30	0.32	0.22	0.21	0.25	0.14	0.16		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page